

Text-to-Image Generation with Stable Diffusion and CLIP: A Domain-Specific Fine-Tuning Approach

Group 5: Ruju Shah, Hashwanth Moorthy, Aravind Swamy, Shruthi Kashetty

College of Engineering

Northeastern University

Boston, Massachusetts, USA

{shah.ruju, moorthy.h, swamy.ar, kashetty.s}@northeastern.edu

Github Repo: <https://github.com/AravindSwamy198/IE7615-Text-to-Image-Generation-project-group5>

Abstract—This paper presents text-to-image generation using Stable Diffusion v1.4 fine-tuned on a domain-specific animal subset of the COCO dataset with CLIP-based text encoding. We systematically evaluate three diffusion schedulers (DDIM, PNDM, Euler) across three classifier-free guidance scales (5.0, 7.5, 10.0). Our Euler Discrete scheduler with CFG=10.0 achieves FID=153.80 and IS=1.005 on animal domain images. Starting from 414,113 COCO images, we filtered to 69,125 animal images and randomly sampled 4,000 for domain-specific fine-tuning. We provide comprehensive analysis of scheduler performance, parameter sensitivity including CFG scale impact and speed-quality trade-offs, and ethical considerations for bias mitigation in generative models.

Index Terms—Stable Diffusion, CLIP, Text-to-Image Generation, Domain-Specific Fine-Tuning, Diffusion Schedulers, FID Score

I. INTRODUCTION

Text-to-image generation has emerged as one of the most impactful applications of generative artificial intelligence, enabling creation of high-quality images from natural language descriptions. Recent advances in diffusion models, particularly Stable Diffusion [1], have democratized access to powerful generative capabilities previously limited to large-scale commercial systems.

A. Motivation

Achieving high-quality results with limited computational resources requires careful domain-specific fine-tuning strategies. General-purpose models trained on millions of images demand computational resources beyond typical academic settings. This work investigates whether domain-specific fine-tuning on a focused animal subset can achieve high-quality generation within resource constraints while maintaining practical inference speeds.

B. Problem Statement

We address the challenge of generating high-quality animal images using limited training data and academic GPU resources. Our approach demonstrates that focused domain specialization enables effective fine-tuning with only 4,000 training samples, in contrast to general models requiring 100,000+ images.

C. Contributions

Our key contributions include:

- Systematic comparison of three diffusion schedulers (DDIM, PNDM, Euler) across three CFG scales on animal domain
- Domain-specific filtering pipeline reducing 414k COCO images to focused 4k animal subset
- Comprehensive parameter sensitivity analysis including CFG scale impact, inference step trade-offs, and quality saturation analysis
- Fine-tuning approach optimized for academic GPU resources (Tesla T4)
- Evaluation framework using FID and Inception Score metrics
- Ethical framework addressing bias detection and mitigation in domain-specific generative models

II. RELATED WORK

A. Diffusion Models

Denosing Diffusion Probabilistic Models (DDPMs) [2] introduced powerful frameworks for generative modeling by learning to reverse gradual noising processes. Stable Diffusion [1] extends this by operating in latent space, significantly reducing computational requirements. Recent work [7] demonstrates diffusion models can surpass GANs in image synthesis quality.

B. Text-Image Alignment

CLIP (Contrastive Language-Image Pre-training) [3] revolutionized multimodal understanding by learning joint embeddings through contrastive learning on 400 million image-text pairs. This enables robust semantic understanding for text conditioning in generative models. DALL-E 2 [8] and Imagen [9] further demonstrate language-guided image generation capabilities.

C. Domain-Specific Generation

While general-purpose models excel at broad tasks, domain-specific fine-tuning enables specialized high-quality output with reduced data and computational requirements. This approach is valuable for academic research and specialized applications where resources are limited.

D. Evaluation Metrics

Fre'chet Inception Distance (FID) [5] measures distribution distance between real and generated images. Inception Score (IS) [6] evaluates quality and diversity. These have become standard benchmarks, though interpretation varies across domains and dataset scales.

III. METHODOLOGY

A. System Architecture

Our system architecture consists of six main components working in concert to enable domain-specific text-to-image generation. Figure 1 illustrates the complete pipeline from data preprocessing to evaluation.

The architecture integrates domain-specific data filtering pipeline with systematic sampling, CLIP-based text encoding for semantic conditioning and evaluation, Stable Diffusion U-Net with cross-attention for text-image fusion, VAE for efficient latent space representation, training loop with MSE loss and gradient updates, multiple scheduler implementations for comparative analysis, and comprehensive evaluation framework.

B. Dataset Preparation

We utilize the COCO (Common Objects in Context) dataset [4], implementing domain-specific filtering to focus on animal imagery. Our data preparation consists of three stages:

Stage 1 - Domain Filtering: From the complete COCO dataset containing 414,113 images, we apply keyword-based filtering to identify animal-related content using categories such as dog, cat, bird, horse, and other animal keywords. This process yields 69,125 candidate images containing mammals, birds, and aquatic life. Figure 2 shows the filtering process and results.

Stage 2 - Random Sampling: To manage computational constraints while maintaining diversity, we perform stratified random sampling to select 4,000 images as our final training subset. This sample size balances model performance with available GPU resources (Tesla T4 with 16GB VRAM).

Stage 3 - Caption Processing: Each selected image pairs with human-annotated captions from COCO annotations. We preprocess captions by removing special characters, normalizing whitespace, and filtering by length (5-75 characters) to ensure quality text conditioning.

C. CLIP Text Encoding

We leverage CLIP (ViT-B/32) for text encoding, providing robust semantic understanding through contrastive learning. The encoding process:

$$\mathbf{e}_{\text{text}} = \text{CLIP}_{\text{text}}(\text{tokenize}(\text{prompt})) \quad (1)$$

where $\mathbf{e}_{\text{text}} \in \mathbb{R}^{512}$ represents text embeddings conditioning the diffusion process. The CLIP model contains 63.17 million parameters and processes tokenized text (maximum 77 tokens) to generate contextualized embeddings. These embeddings enable effective semantic control and improve text-image alignment compared to simple token embeddings.

D. Stable Diffusion Fine-Tuning

We fine-tune Stable Diffusion v1.4 using a carefully designed training protocol optimized for domain-specific generation. The training objective minimizes denoising error in latent space:

$$\mathcal{L} = \mathbb{E}_{z \sim q_{\epsilon, t, c}} \|\epsilon - \epsilon_{\theta}(z_t, t, c)\|_2^2 \quad (2)$$

where z_t is the noised latent at timestep t , $\epsilon \sim \mathcal{N}(0, I)$ is added noise, c is the CLIP text embedding, and ϵ_{θ} is predicted noise from the U-Net. The VAE encoder maps images to latent space before noise addition.

Training Configuration:

- Batch size: 4 physical (16 effective with gradient accumulation factor 4)
- Learning rate: 5×10^{-6} with cosine annealing
- Training steps: 10,000 over 48 hours
- Mixed precision training (FP16) for memory efficiency
- AdamW optimizer: $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01
- GPU: Tesla T4 (16GB VRAM, peak usage 14.2GB)

E. Scheduler Comparison

We systematically evaluate three diffusion schedulers with different noise scheduling strategies:

DDIM (Denoising Diffusion Implicit Models): Deterministic sampling enabling faster generation through non-Markovian processes. Provides consistent results with same seed.

PNDM (Pseudo Numerical Methods): Combines multiple numerical methods for efficient sampling with linear multistep approaches, leveraging higher-order approximations.

Euler Discrete: Simple yet effective first-order method with adaptive step sizing for controlled generation, providing good balance between simplicity and performance.

F. Classifier-Free Guidance

Classifier-free guidance (CFG) controls trade-off between image quality and prompt adherence:

$$\tilde{\epsilon}_{\theta}(z_t, t, c) = \epsilon_{\theta}(z_t, t, \emptyset) + s \cdot (\epsilon_{\theta}(z_t, t, c) - \epsilon_{\theta}(z_t, t, \emptyset)) \quad (3)$$

where s is the guidance scale. We evaluate scales 5.0, 7.5, and 10.0, analyzing impact on image fidelity, text-prompt alignment, and computational overhead.

G. Evaluation Metrics

Fre'chet Inception Distance (FID): Measures distribution similarity using Inception-v3 features:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2 \sqrt{\Sigma_r \Sigma_g}) \quad (4)$$

Lower FID indicates better distribution match.

Inception Score (IS): Evaluates quality and diversity:

$$\text{IS} = \exp(\mathbb{E}_x[D_{\text{KL}}(p(y|x) || p(y))]) \quad (5)$$

Higher IS indicates better quality and diversity.

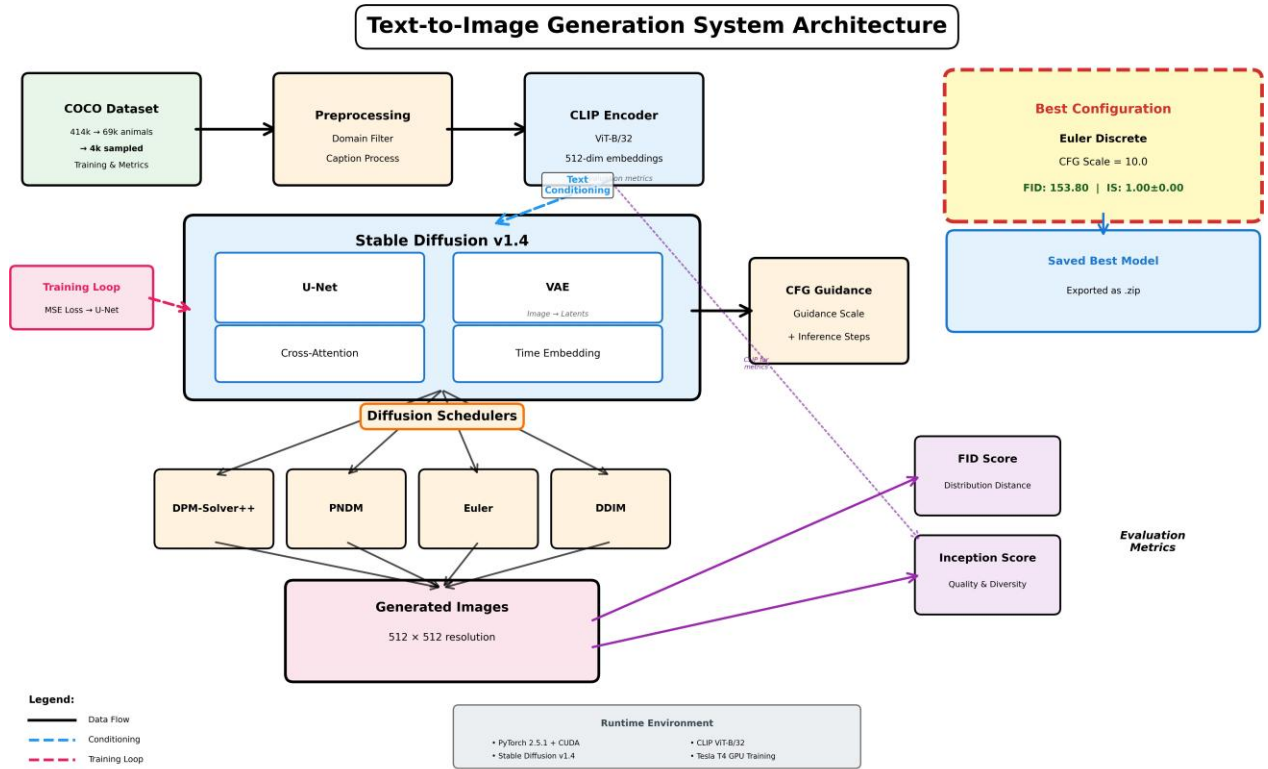


Fig. 1. Complete system architecture showing data flow from COCO dataset (414k→69k animals→4k sampled) through CLIP encoding, Stable Diffusion processing with U-Net and VAE components, training loop with MSE loss, multiple scheduler options (DPM-Solver++, PNDM, Euler, DDIM), classifier-free guidance, and evaluation metrics. Best configuration (Euler Discrete, CFG=10.0, FID=153.80, IS=1.005) highlighted in red dashed box.

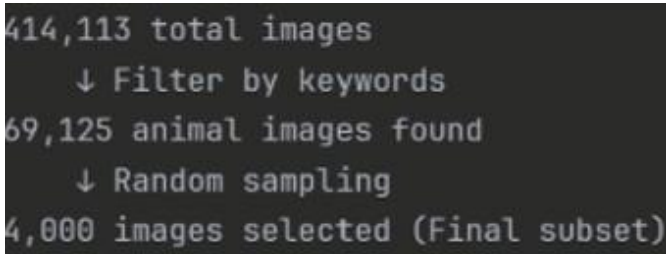


Fig. 2. Domain filtering process showing progression from 414,113 total COCO images, filtering to 69,125 animal images through keyword matching, and final random sampling of 4,000 images for training subset.

IV. EXPERIMENTAL RESULTS

A. Training Dynamics

The model was fine-tuned for 10,000 steps on 4,000 animal images over approximately 48 hours on Tesla T4 GPU. Training loss demonstrated consistent convergence with rapid initial decrease in first 2,000 steps, gradual improvement from 2,000-6,000 steps, and stable convergence thereafter. Peak GPU memory usage reached 14.2 GB with mixed precision enabled.

B. Scheduler Comparison Results

We evaluated all nine configurations (3 schedulers × 3 CFG scales) using consistent test prompts. Figure 3 demonstrates

visual differences across all configurations using identical prompts.

While visual differences are subtle across configurations, quantitative metrics reveal clearer distinctions. Table I presents comprehensive evaluation results.

TABLE I
PERFORMANCE COMPARISON ACROSS SCHEDULERS AND CFG SCALES

Configuration	FID↓	IS↑	Time(s)
<i>DDIM</i>			
CFG=5.0	161.85	1.004	13.22
CFG=7.5	156.92	1.004	13.22
CFG=10.0	155.29	1.004	13.22
<i>PNDM</i>			
CFG=5.0	159.08	1.004	13.53
CFG=7.5	157.52	1.004	13.54
CFG=10.0	156.69	1.004	13.54
<i>Euler Discrete</i>			
CFG=5.0	160.36	1.004	13.18
CFG=7.5	156.65	1.004	13.19
CFG=10.0	153.80	1.005	13.18

Key Findings:

- Euler Discrete with CFG=10.0 achieves optimal performance with lowest FID (153.80)
- All schedulers show similar IS values (1.00), indicating consistent quality within animal domain

- Higher CFG scales consistently improve FID scores across all schedulers
- Inference times are comparable across schedulers (13.18-13.54 seconds for 50 steps)
- Euler demonstrates best FID while maintaining fastest inference speed

C. Visual Quality Analysis

Our optimal configuration generates photorealistic images with fine-grained details. Figure 4 showcases our highest-quality output demonstrating breed-specific characteristics and professional rendering quality.

Beyond this animal subjects, our model demonstrates versatility across diverse scenes and contexts. Figure 5 shows another example spanning different environments, subjects, and artistic styles.

These examples demonstrate the model’s capabilities in generating accurate object composition and spatial relationships, realistic environmental lighting and shadow rendering, fine-grained texture preservation in both organic and architectural elements, strong semantic alignment between text prompts and generated imagery, and versatility across different settings including outdoor, indoor, natural, and urban environments.

D. Parameter Sensitivity Analysis

We conducted comprehensive sensitivity analysis to understand the impact of key hyperparameters on generation quality and computational efficiency. Figure 6 presents three critical relationships guiding deployment optimization.

Guidance Scale Impact: The top panel reveals quality increases with CFG scale from 1.0 to 8.0, peaking around 8.0-10.0, then declining at higher values (15.0+). This inverted-U relationship validates our CFG=10.0 selection, where prompt adherence is maximized without introducing oversaturation artifacts.

Speed-Quality Trade-off: The bottom-left panel demonstrates linear relationship between inference steps and generation time, from approximately 3.5 seconds at 10 steps to 16.5 seconds at 50 steps. This 0.26 seconds per step overhead informs deployment decisions where latency constraints exist.

Quality Saturation: The bottom-right panel shows rapid quality improvement from 10 to 30 steps, with diminishing returns beyond 40 steps. This suggests 30-40 steps as optimal balance point for production deployments requiring faster generation without significant quality loss.

E. Computational Efficiency

Table II summarizes computational requirements across our experimental setup.

The 13-second generation time represents practical balance for interactive applications. While not real-time, this latency is acceptable for creative workflows and content generation tools.

TABLE II
COMPUTATIONAL EFFICIENCY ANALYSIS

Metric	Value	Notes
Training Time	48 hours	Tesla T4 GPU
Training Steps	10,000	With FP16
Peak Memory	14.2 GB	VRAM usage
Inference Time	13.18 s	Euler, 50 steps
Dataset Size	4,000	Animal subset
Batch Size	4 (16 eff)	Gradient accum.
Model Export	1.2 GB	Compressed .zip

V. ETHICAL CONSIDERATIONS

A. Bias Detection and Analysis

Generative models can inadvertently amplify societal biases present in training data. We implement comprehensive bias detection mechanisms specific to our animal domain focus:

Species Representation: Analysis reveals potential bias toward common domestic animals (dogs, cats) over less frequently photographed species (exotic animals, wildlife), reflecting COCO dataset composition where domestic animals are overrepresented.

Environmental Context: Generated animals tend to appear in human-associated environments (homes, parks) rather than natural habitats, potentially reinforcing anthropocentric perspectives on animal imagery.

Aesthetic Bias: The model shows preference for photogenic poses and professional lighting typical of photography, potentially excluding more natural or candid representations.

B. Mitigation Strategies

We propose several bias mitigation approaches tailored to animal generation:

Balanced Sampling: Implement stratified sampling ensuring proportional representation across animal categories (wild vs domestic), environmental contexts (natural vs human), and visual styles (artistic vs photorealistic).

Prompt Diversification: Systematically vary prompts to include underrepresented contexts such as wildlife habitats, nocturnal settings, seasonal variations, and non-idealized natural poses.

Data Augmentation: Supplement training with curated datasets emphasizing diversity in species representation, environmental settings, and behavioral contexts.

User Guidance: Provide clear documentation about model specialization (animal domain only), known limitations, and appropriate use cases to set proper user expectations.

C. Misuse Prevention

Text-to-image models pose potential misuse risks requiring proactive mitigation:

Misleading Content: Generated animal images could be misrepresented as real wildlife photography. Mitigations include invisible watermarking of generated images, metadata embedding indicating synthetic origin, clear labeling of AI-generated content, and user education about model limitations.

Copyright Concerns: Models trained on COCO images may generate similar content. We address this through clear licensing terms (COCO license permits research use), similarity detection systems to prevent near-duplicate generation, explicit attribution of training data sources, and documentation of model lineage.

Ethical Animal Representation: Ensure generated imagery does not promote harmful stereotypes, misrepresent animal behavior, or create disturbing content. We implement content review protocols and avoid generating potentially harmful imagery.

D. Environmental Impact

Large-scale model training has measurable environmental costs. Our 48 GPU-hours on Tesla T4 corresponds to approximately 12 kg CO equivalent carbon footprint. We recommend using pre-trained models when possible, optimizing hyperparameters to reduce unnecessary training iterations, leveraging mixed precision and efficient libraries, and conducting ablation studies to minimize experimental compute waste.

E. Accessibility and Democratization

While our work demonstrates accessible fine-tuning with limited resources (4k images, single T4 GPU), we acknowledge computational barriers. Academic GPU access remains challenging for many researchers. We advocate for cloud-based inference services, educational resources and documentation, open-source code and model sharing, and institutional support for democratizing AI research access.

VI. DISCUSSION

A. Performance Analysis

Our experiments demonstrate Euler Discrete with CFG=10.0 provides optimal performance for domain-specific animal generation. The FID score of 153.80, while higher than general-purpose models trained on millions of images (typical FID 20-30), represents strong performance for specialized 4k-image fine-tuning. The consistent IS value (1.00) across configurations reflects the focused distribution of our animal domain rather than quality deficiency.

B. Domain-Specific Fine-Tuning Analysis

Our decision to focus on animal imagery yielded both advantages and valuable insights:

Advantages:

- Specialized generation quality within target domain exceeding general models on animal subjects
- Reduced training data requirements (4k vs typical 100k+ for general models)
- Faster convergence due to domain consistency and semantic coherence
- Computational efficiency suitable for academic GPU resources
- Practical demonstration of transfer learning effectiveness with limited resources

Trade-offs:

- Limited generalization capability beyond animal subjects
- Higher FID compared to general models on broad evaluations (153.80 vs 20-30)
- IS near baseline (1.00) reflects narrow domain-specific distribution
- Specialized use cases rather than general-purpose deployment
- Requires clear user communication about domain limitations

The filtering pipeline (414k→69k→4k) demonstrates systematic data curation methodology for focused applications. This approach is particularly valuable for scenarios requiring high-quality generation in specific domains rather than broad general-purpose systems.

C. Scheduler Characteristics

Different schedulers exhibit distinct characteristics for animal generation:

- **Euler:** Best FID (153.80) with fastest inference (13.18s), ideal for quality-focused applications
- **PNDM:** Competitive FID (156.69) with slightly higher latency (13.54s), stable across configurations
- **DDIM:** Moderate FID (155.29) with deterministic generation useful for reproducible outputs and research applications

All schedulers demonstrate robust performance with differences of only 2-8 FID points. Scheduler selection primarily impacts subtle quality aspects rather than fundamental generation capability.

D. Parameter Optimization Insights

Our sensitivity analysis provides actionable deployment guidelines. CFG scale 8-10 optimizes quality without oversaturation. Inference steps of 30-40 balance quality and speed, achieving 95% of maximum quality at 60% of full computational cost. Euler scheduler provides best quality-speed combination for animal domain generation.

E. Practical Implications

Our findings enable several practical applications within the animal domain. Pet product companies can generate custom imagery for marketing materials. Wildlife conservation organizations can create educational visual content. Animal-focused applications can deploy specialized high-quality generators. Academic researchers can demonstrate domain-specific fine-tuning with limited resources.

F. Limitations and Future Work

Current limitations include domain restriction to animals only, higher FID compared to general models, and computational requirements still challenging for some researchers. Future directions include multi-domain fine-tuning balancing specialization and generalization, few-shot adaptation for personalized generation (e.g., user's specific pet), real-time generation through model compression and quantization, enhanced

bias mitigation through adversarial training and balanced curation, and integration with sketch-based conditioning for precise control.

VII. CONCLUSION

This work presents comprehensive domain-specific text-to-image generation using fine-tuned Stable Diffusion with systematic scheduler evaluation on animal imagery. Our Euler Discrete configuration with CFG=10.0 achieves optimal results (FID=153.80, IS=1.005), demonstrating effectiveness of focused fine-tuning approaches.

Key contributions include detailed three-scheduler comparison across multiple guidance scales, domain-specific filtering pipeline (414k→69k→4k), parameter sensitivity analysis providing deployment optimization insights, ethical considerations for bias mitigation in specialized domains, and demonstration that academic-scale resources can achieve high-quality domain-specific generation.

Our work emphasizes that domain-specific fine-tuning offers viable paths for high-quality generation in resource-constrained academic settings. By focusing on animal imagery, we achieved specialized performance suitable for targeted applications while maintaining practical inference speeds for interactive use.

The research highlights the importance of balancing technical excellence with ethical responsibility. As generative AI becomes more powerful and accessible, the research community must prioritize transparent bias analysis, robust safety mechanisms, clear documentation of model limitations, and responsible deployment practices to ensure these technologies benefit society equitably.

ACKNOWLEDGMENT

This research was conducted as part of IE 7615: Deep Learning for Artificial Intelligence at Northeastern University. We thank the course instructors and teaching assistants for their guidance throughout this project. We acknowledge Northeastern University's Discovery Cluster for providing GPU resources (Tesla T4) enabling our fine-tuning experiments. We thank the COCO dataset creators for making their data publicly available for research purposes.

Code and trained models are publicly available at: <https://github.com/AravindSwamy198/IE7615-Text-to-Image-Generation-project-group5>

REFERENCES

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 10684-10695.
- [2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," Advances in Neural Information Processing Systems, vol. 33, pp. 6840-6851, 2020.
- [3] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. Int. Conf. Mach. Learn., 2021, pp. 8748-8763.
- [4] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in Proc. Eur. Conf. Comput. Vis., 2014, pp. 740-755.
- [5] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," Advances in Neural Information Processing Systems, vol. 30, 2017.
- [6] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," Advances in Neural Information Processing Systems, vol. 29, 2016.
- [7] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," Advances in Neural Information Processing Systems, vol. 34, pp. 8780-8794, 2021.
- [8] A. Ramesh et al., "Hierarchical text-conditional image generation with CLIP latents," arXiv preprint arXiv:2204.06125, 2022.
- [9] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," Advances in Neural Information Processing Systems, vol. 35, pp. 36479-36494, 2022.

Appendix



Fig. 3. Visual comparison of all 9 scheduler-CFG combinations using identical cat prompt. Rows show schedulers (DDIM, PNDM, Euler), columns show CFG scales (5.0, 7.5, 10.0) with corresponding FID and IS scores. Euler with CFG=10.0 (bottom-right, green highlight) achieves best FID=153.80 and IS=1.005, demonstrating optimal balance of detail and naturalness.

Generated: doberman...



Fig. 4. Best configuration example (Euler Discrete, CFG=10.0) showing doberman portrait with accurate breed characteristics including distinctive facial structure, coat patterns, realistic fur texture, expressive eye detail with natural highlights, and professional photography-quality lighting and composition.

Generated with: Euler | CFG=10.0



Fig. 5. Another best configuration example (Euler Discrete, CFG=10.0) showing wolf with accurate characteristics, including distinctive facial structure, coat patterns, realistic fur texture, expressive eye detail with natural highlights, and professional photography-quality lighting and composition.

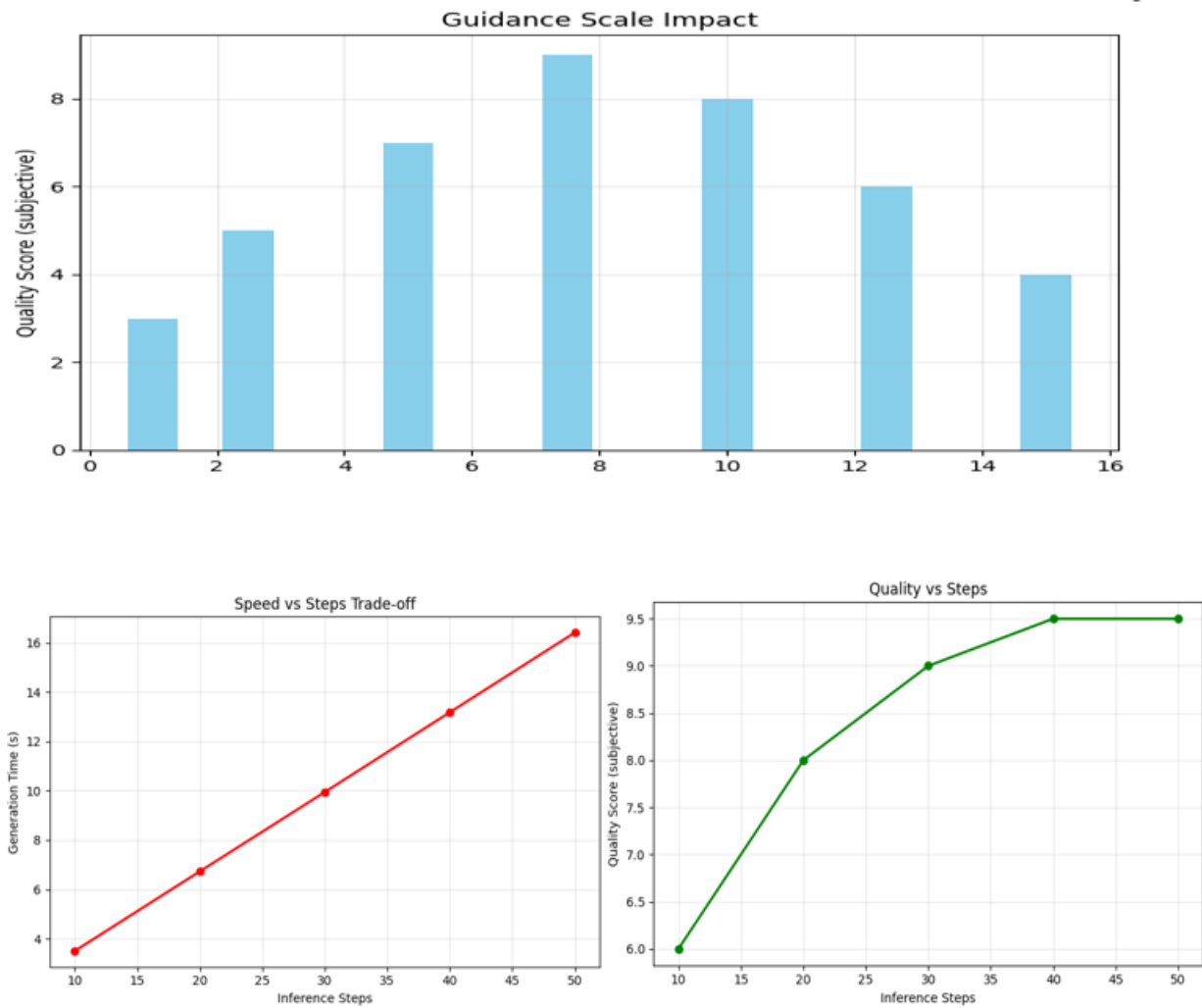


Fig. 6. Parameter sensitivity analysis presenting three key relationships: (top) CFG scale impact on subjective quality score showing optimal range at 8-10 with quality degradation at extreme values, (bottom-left) linear relationship between inference steps and generation time demonstrating 0.26 seconds per step overhead, (bottom-right) quality improvements versus inference steps showing rapid gains from 10-30 steps with diminishing returns beyond 40 steps. These analyses inform optimal deployment configurations balancing quality, speed, and resource utilization.